

Brainstorming & Idea Prioritization

OptiCrop — Smart Agricultural Production Optimization Engine

Raw Ideas Generated

- A mobile/web app where farmers manually look up crop suitability from a static reference table.
- A chatbot that answers farming questions using general agricultural knowledge.
- A machine learning model that predicts the best crop from soil (N, P, K, pH) and climate (temperature, humidity, rainfall) inputs.
- A weather-only crop calendar that suggests planting windows by season.
- A community forum where farmers crowdsource crop advice from each other.
- An IoT sensor kit that measures soil values directly and feeds them into a prediction engine.

Prioritization Criteria

- Feasibility within a 7-day solo build (no hardware dependency, achievable with available datasets).
- Accuracy / reliability of the recommendation (objective, data-driven vs. opinion-based).
- Real-world impact on farmer decision-making and resource efficiency.
- Extensibility toward research and policy use cases (Scenario 3 in the original brief).

Prioritization Outcome

The ML-based crop recommendation engine (idea 3) scored highest on feasibility and impact: it does not depend on hardware (unlike the IoT idea), is objectively data-driven (unlike the forum or static-table ideas), and directly supports all three target user groups — farmers, researchers, and policymakers. This idea was selected and developed into OptiCrop.

Selected Idea: OptiCrop

A web application that takes soil nutrient levels (N, P, K), temperature, humidity, pH, and rainfall as input, and uses a trained classification model to recommend the most suitable crop for those exact conditions — replacing guesswork with a measurable, repeatable recommendation.